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Geological Society of America Bulletin 1990;102;1517-1525
doi: 10.1130/0016-7606(1990)102<1517:POTCRD>2.3.CO;2

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Paleomagnetism of the Cambrian Royer Dolomite and Pennsylvanian Collings Ranch Conglomerate, southern Oklahoma: An early Paleozoic magnetization and nonpervasive remagnetization by weathering

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ABSTRACT

The Cambrian Royer Dolomite in the Arbuckle Mountains, Oklahoma, contains an early Paleozoic magnetization and a nonpervasive, late Paleozoic remagnetization caused by meteoric fluids. The Royer was uplifted and underwent karst conditions, and eroded clasts were deposited in the Collings Ranch Conglomerate during the Pennsylvanian. An early Paleozoic, depositional or chemical remanent magnetization (Dec. = 109°, Inc. = 10°, tilt corrected) residing in magnetite is found in gray-brown, ferroan dolomite. A Pennsylvanian-Permian chemical remanent magnetization (CRM) residing in hematite (Dec. = 147°, Inc. = 4°, *in situ*) is associated with dedolomitized rocks in an alteration zone around and below the karst features. Clasts of the Royer in the Pennsylvanian Collings Ranch Conglomerate, correlated with the parent rock by petrographic, isotopic, and rock magnetic methods, also contain two magnetizations. The clast centers contain a predepositional magnetization, whereas the clast margins contain a postdepositional CRM (Dec. = 152°, Inc. = 4°, *in situ*). Conglomerate tests using these magnetization components constrain the timing of remanence acquisition in the Royer Dolomite.

Field relations and the results of oxygen and carbon stable-isotope analyses indicate that the dedolomitization events were caused by near-surface, meteoric fluids that altered the Royer around the margin of the karst dissolution caves and the outer rims of Royer Dolomite clasts. Remagnetization was not pervasive, and an early Paleozoic magnetization is preserved in the dolomite below the zone affected by weathering.

INTRODUCTION

Many rocks have magnetizations that are younger than the depositional age of the unit. In particular, many early Paleozoic units were remagnetized in the late Paleozoic (Pennsylvanian to Permian). Because late Paleozoic remagnetizations are so numerous, large-scale, limited-duration events are commonly assumed to have been responsible. An early explanation for this late Paleozoic event was that severe weathering caused hematite authigenesis and remagnetization when North America was located in tropical latitudes (Creer, 1968). Another hypothesis relies on thermoviscous remagnetization due to cooling from elevated burial temperatures (Kent, 1985). A third large-scale event involves the effect of "orogenic fluids" driven through a large mass of rock as a result of continental collision (for example, Oliver, 1986).

In order to better understand the origin of late Paleozoic remagnetizations, it is important that individual magnetizations be related to specific processes. In this study, we relate a late Paleozoic chemical remanent magnetization (CRM) in the Cambrian Royer Dolomite to a specific, nonpervasive, remagnetizing event caused by meteoric fluids associated with weathering. In addition, an early Paleozoic magnetization in the Royer Dolomite that was not affected by weathering was identified. Studies of clasts of the Royer in the Upper Pennsylvanian Collings Ranch Conglomerate also constrain the time of remanence acquisition for both components.

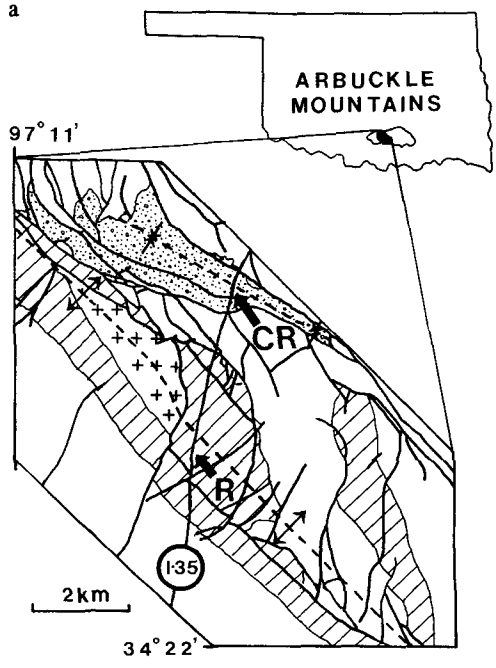


Figure 1. Simplified general geologic and location map. Sampling sites marked by R (Royer Dolomite) and CR (Collings Ranch Conglomerate). Dashed lines mark axes of the Arbuckle anticline and syncline in the Collings Ranch Conglomerate. Crosses, Cambrian Colbert Rhyolite and Timbered Hills Group; diagonal lines, Cambrian Lower Arbuckle Group including Royer Dolomite; no pattern, Ordovician units; conglomerate pattern, Pennsylvanian Collings Ranch Conglomerate. Thick lines are fault contacts, and thin lines are unit contacts.

Geological Society of America Bulletin, v. 102, p. 1517–1525, 13 figs., 2 tables, November 1990.

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GEOLOGIC SETTING AND PREVIOUS WORK

The Royer Dolomite and Collings Ranch Conglomerate are exposed in the Arbuckle Mountains in south-central Oklahoma (Fig. 1). The Royer Dolomite is a stratiform dolomite within the Upper Cambrian Fort Sill Formation (Sargent, 1969; Ham, 1978) and formed as a result of early dolomitization of shallow-marine limestones in the southern Oklahoma aulacogen. Sargent (1969) identified at least two stages of dolomitization in the Arbuckle Mountains. One episode probably occurred relatively early after deposition and resulted in a "stratiform" or roughly bedding-parallel dolomite such as the Royer. A later dolomitizing event is related to Pennsylvanian-age faults and affected other limestone units and some of the older stratiform dolomite.

During Middle to Late Pennsylvanian time, the Arbuckle anticline formed, and the region was uplifted, deformed, and eroded. Karst dissolution, related to uplift during the Pennsylvanian, affected part of the Royer Dolomite (Ham, 1978). The Royer Dolomite is normally gray to brown in color, but surrounding the karst features is a pink to red halo that averages about 50 cm in thickness (Fig. 2a). The paleocaves are now filled with weathered flowstone, clastic sediment, and carbonate cements.

During uplift of the Arbuckle Mountains, a series of fault-block basins formed. These grabens contain cobbles of Paleozoic rocks eroded from the adjacent highlands. During one of these erosional/depositional stages, the Collings Ranch Conglomerate (Late Pennsylvanian) was deposited in a graben-like basin about 1 km wide and 8 km long (Ham, 1954; Wickham and Denison, 1978; Donovan and Heinlen, 1988). The unit consists of limestone and dolomite clasts as much as 1 m in size, derived from early to middle Paleozoic units (Fig. 2b). The conglomerate is deformed into an asymmetric syncline the limbs of which dip as much as 75°. Clasts of the Royer Dolomite can be identified at the outcrop. They are light pink to dark red in color and composed of medium-crystalline dolomite and dedolomite (calcite).

Previous paleomagnetic studies in the Arbuckle Mountains documented that both dedolomitized and Liesegang-banded lithologies contain Late Pennsylvanian to Permian CRM's residing in authigenic hematite (Elmore and others, 1985; Cochran and Elmore, 1987). A diagenetic origin for these CRM's contrasts with a more regional interpretation of a viscous partial thermal magnetization (VpTRM) proposed by Steiner (1973) for the same rocks in the Arbuckle Mountains. The presence of VpTRM's in the region is unlikely, however, given a low conodont alteration index for the rocks (Elmore and others, 1985).

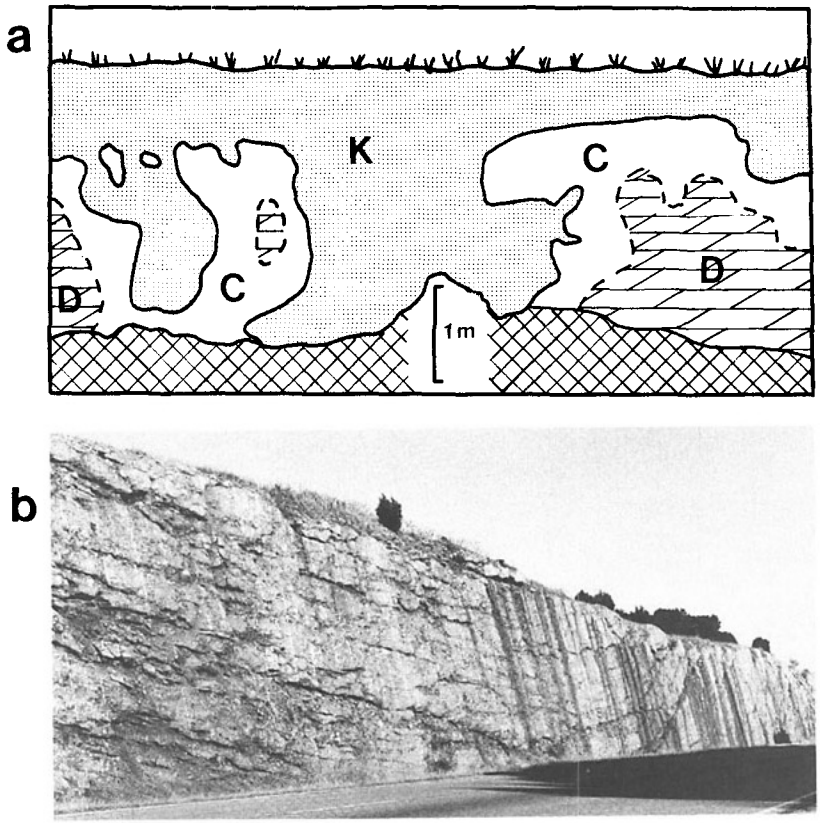


Figure 2. Field characteristics of sample sites. (a) Line drawing of karst feature in the Royer Dolomite, traced from outcrop photographs. Outcrop is located at intersection of Interstate 35 and Highway 77 at the R location in Figure 1. K is clastic karst fill and alluvium, C is red dedolomite (calcite) halo, D is gray to brown Royer Dolomite, and diagonally lined area is covered. (b) Photograph of an outcrop of Collings Ranch Conglomerate on the west side of the north-bound lane of Interstate 35, location CR in Figure 1, showing synclinal fold.

SAMPLING AND LABORATORY TECHNIQUES

Gray to brown dolomite and reddened halos surrounding the paleokarst surfaces were sampled in the Royer Dolomite. Samples were taken from the gray-brown dolomite near the crest and the southern flank of the Arbuckle anticline. The reddened dolomite associated with the karst was found only near the crest of the anticline. Clasts of the Royer Dolomite within the Collings Ranch Conglomerate fall into three groups. These include clasts of a uniform pink to gray color, those with a uniform dark red color that are commonly porous, and clasts with pink to gray centers and dark red rims. Samples of each clast type were taken from the axis and each limb of the syncline.

In all, 245 cores (2.5-cm diameter) were drilled and oriented with a compass and clinometer from outcrops along Interstate 35 and State Highway 77 in the Arbuckle Mountains, southern Oklahoma (Fig. 1). Standard specimens of 2.2-cm length were cut from the cores. The remanent magnetization was measured with a United Scientific, three-axis, cryogenic magnetometer located in a magnetically shielded laboratory. Progressive thermal and alternating field (AF) demagnetization was used to isolate and identify the components of the natural remanent magnetization (NRM). Specimens were heated at constant temperature steps (100, 200, 300, 350, 400, 450, 500, 540, 580, 600, 640, 680, 700 °C) for 30 min, then cooled to room temperature in a <2 nT field. AF demagnetization was by external alternating fields of 10, 30, 50, 75, 90, 125, and 150 mT. Orthogonal projections (Zijderveld, 1967) of the magnetization directions and intensity at each decay step were plotted and the remanence directions calculated

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by principal-component analysis (Kirschvink, 1980). Further calculations were limited to those principal components with a maximum angle of deviation of less than 6.0°.

Coercivity spectrum analysis was used to aid in identification of magnetic mineralogies (Dunlop, 1972; Lowrie and Heller, 1982). Specimens were subjected in a stepwise fashion to an increasing isothermal remanent magnetization (IRM) of as much as 0.8 T and then thermally demagnetized.

Chips from the drilled cores and hand samples from similar lithologies were used for petrographic and isotopic analyses. Thin sections were examined with transmitted and reflected light and cathodoluminescence microscopy. Fractured or polished and etched chips were examined with a scanning electron microscope (SEM) with energy dispersive analysis capability.

Representative samples were removed and powdered for oxygen and carbon stable-isotope analysis. The powders were reacted with 100% phosphoric acid, and carbon- and oxygen-isotope ratios of the CO_2 gas measured with a Finnigan Delta E mass spectrometer. Stable isotope data are presented below as

$$\delta N_E{}^0/_{00} = [R_{SAMPLE}/R_{STANDARD} - 1] \times 10^3,$$

where N is the heavier stable isotope of the element E, and R is the abundance ratio of the heavy to light isotopes of the element. The standard used for ^{13}C and ^{18}O was the Chicago PDB. For routine measurement, samples were analyzed versus a laboratory substandard tank of pure CO_2 gas calibrated against NBS reference materials.

RESULTS AND INTERPRETATION

Petrographic and Isotopic Analyses

The Royer Dolomite is composed of fine-grained, xenotopic dolomite with scattered, coarse-grained, euhedral rhombs. Staining shows that the fine-grained dolomite is uniformly ferroan, and coarse crystals are zoned in iron content. The dolomite has a uniform, dull red cathodoluminescence. Scattered, very fine-grained pyrite crystals are present. Magnetic extracts from crushed and partially dissolved dolomite contain spheres (Fig. 3a) with iron as the only detectable element, based on energy dispersive analysis (EDA). The shape and surface morphology of the spheres are consistent with previously reported authigenic magnetite (McCabe and others, 1983; Elmore and others, 1987).

In thin section, the reddened zones around the karst feature consist predominantly of micritic to finely crystalline mosaic calcite with scattered, coarse-grained, rhomb-shaped pseudomorphs filled with coarse-grained, equant spar. Iron oxides are concentrated along irregular lines in the calcite and within and at the outer margins of the rhombic pseudomorphs (Fig. 3b). In contrast to the dolomite, most of the calcite luminesces bright orange.

The transitional zone between calcite and dolomite exhibits two types of replacement textures. In fine-grained dolomite, the calcite is also fine grained with scattered, preserved areas of dolomite. Some anastomosing calcite veins invade areas of fine-grained dolomite. In the larger dolomite crystals, equant spar either totally replaces the dolomite or preferentially replaces certain growth zones in the dolomite crystals.

Clasts of the Royer Dolomite in the Collings Ranch Conglomerate show analogous petrographic characteristics to the parent Royer. Many clast centers are composed of pink to gray ferroan dolomite. The completely red clasts and the red clast rims contain calcite replacement textures that are the same as those previously described for the calcitized karst halo. A pink transition zone commonly separates the clast rims and centers.

Stable-isotope values of carbon and oxygen determined for the gray dolomite and reddened calcitized dolomite in the Royer and for Royer clasts in the Collings Ranch Conglomerate fall into two groups corresponding to the two petrographic categories. The gray Royer Dolomite and clast centers have similar $\delta^{18}O$ and $\delta^{13}C$ values, and the calcitized dolomites from the karst halo and clast rims also have similar isotopic values (Fig. 4).

Interpretation

In the Royer Dolomite, petrographic results show that the fine-grained dolomite has been

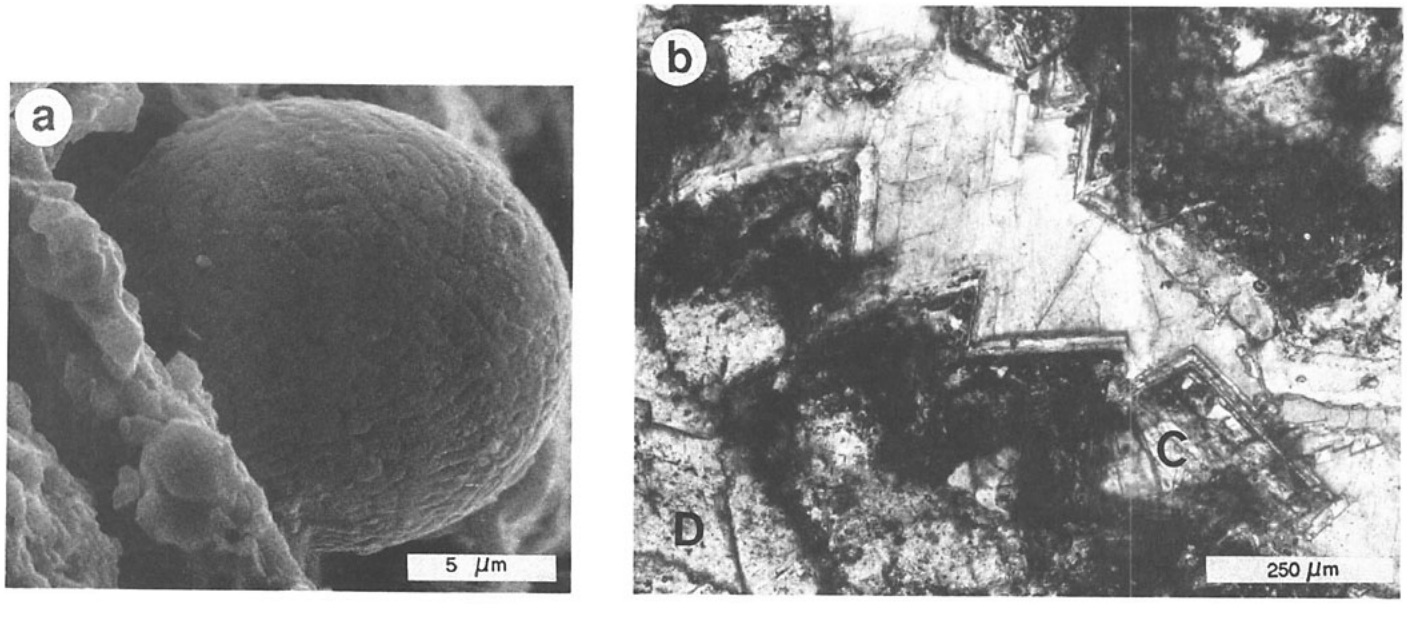


Figure 3. Petrographic characteristics of samples. (a) SEM photomicrograph of a sphere from a magnetic extract of the Royer Dolomite. Iron was the only element detected in the sphere by energy dispersive analysis. (b) Photomicrograph showing calcite (C) that has replaced dolomite (D) and the associated, dark iron oxides.

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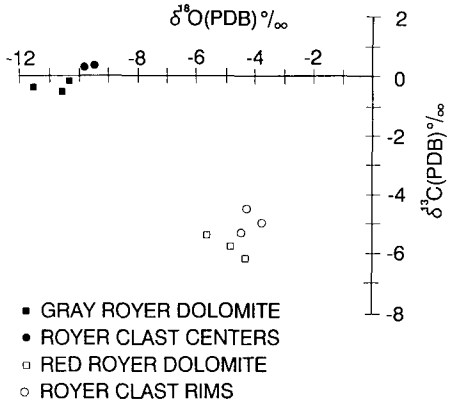


Figure 4. Carbon and oxygen stable-isotope compositions (PDB) of dolomite and calcite. Filled squares are Royer Dolomite, and filled circles are clast centers from Royer in the Collings Ranch. Open squares are calcites (dedolomite) from karst halo, and open circles are calcites from clast rims.

TABLE 1. MEAN MAGNETIZATION DIRECTIONS AND ASSOCIATED APPARENT POLE POSITIONS FROM THE ROYER DOLOMITE

Site	n/n_0	Dec.°	Inc.°	k	α_{95}°	Lat. °N	Long. °E	dp	dm
Gray-brown Royer Dolomite									
no tilt	32/36	110.6	14.1	27.0	4.8	12.6	154.9	2.5	4.9
tilted	32/36	109.2	10.5	33.3	4.3	12.6	157.3*	2.2	4.4
Reddened Royer Dolomite from karst halo									
no tilt	17/25	147.1	3.8	95.9	3.7	42.3	130.1*	1.8	3.7
tilted	17/25	147.3	13.8	95.9	3.7	38.3	126.0*	1.9	3.7
Karst fill material in Royer Dolomite									
no tilt	6/6	9.5	56.2	113.4	6.3	81.9	332.9	6.6	9.1
tilted	6/6	0.2	49.1	113.4	6.3	85.6	80.4	5.5	8.4

Note: Dec. and Inc. are declination and inclination, respectively, in degrees; n/n_0 is the ratio of specimens used in the statistical analysis to the number of specimens studied (one specimen per sample); k is the precision parameter associated with the specimen or site mean (Fisher, 1953); α_{95} is the radius in degrees of the cone of 95% confidence of the mean; Lat. and Long. are the respective latitude and longitude of the apparent pole; dp and dm are the semi-major and semi-minor axes in degrees, respectively, of the oval of 95% confidence around the poles associated with the paleomagnetic directions; *indicates that the pole is the antipodal equivalent of the calculated pole position.

locally replaced by calcite (dedolomite) and reddened, owing to hematite authigenesis. Royer clasts in the Collings Ranch Conglomerate show similar replacement textures. When ferroan dolomite is replaced by calcite in an oxidizing setting, ferrous ions are released, oxidized, and form oxides that co-precipitate with calcite (Evamy, 1969; Katz, 1971). Mass-balance calculations show that the ferroan dolomite can easily account for the amount of hematite in the reddened Royer, although a small amount of the hematite may form by oxidation of minor pyrite in the precursor rock. Dedolomitization has been demonstrated in the surface (Evamy, 1967; Al-Hashimi and Hemingway, 1973; Frank, 1981) and subsurface (Budai and others, 1984; Theriault and Hutcheon, 1987) settings. Karst formation is a near-surface process mediated by meteoric fluids (Esteban and Klappa, 1983), and stratigraphic evidence suggests that karst formation in the Royer took place after uplift of the Arbuckle Mountains. This suggests that the dedolomitization in the Royer occurred in a near-surface setting.

Similar stable-isotope values for carbon and oxygen (Fig. 4) suggest that the dolomite in the center of the clasts is similar to the dolomite in the Royer and has probably not been differentially affected by recrystallizing fluids. The grouping of the stable-isotope values for dedolomite (calcite) for the karst halo and clast rims also suggests that a similar fluid was responsible for dedolomitization in each case.

The stable-isotope values for oxygen also suggest that the calcites precipitated from meteoric waters. The Arbuckle Mountains were at low latitudes during the Pennsylvanian-Permian (Irving and Irving, 1982). The $\delta^{18}O$ values of the calcites (dedolomites) are consistent with expected values for calcite in equilibrium with low-latitude precipitation (Craig and Gordon, 1965) and near-surface temperatures (O'Neil and others, 1969).

Reddening of the outer surfaces of clasts of Royer Dolomite in the Collings Ranch Conglomerate suggests that the alteration took place after individual clasts were eroded and deposited. Petrographic and isotopic evidence supports the use of a conglomerate test (described

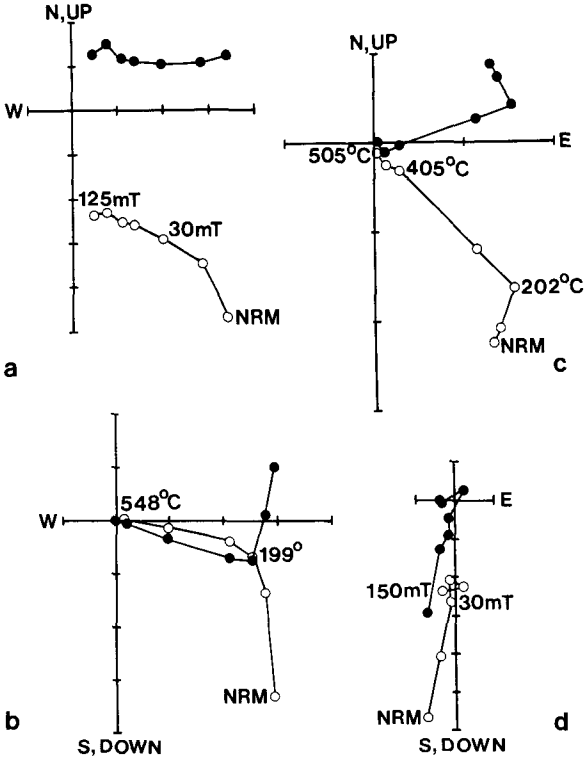


Figure 5. Orthogonal projections of demagnetization data from representative specimens. (a) AF decay of magnetization in a gray Royer specimen, showing removal of eastern and shallow to moderate remnant direction. (b) Thermal decay of magnetization in a gray Royer sample, showing initial low-temperature removal of northern and steep component followed by eastern and moderate to shallow component with decay to origin. (c) Thermal decay of the magnetization in a specimen from the gray dolomite clast center, showing two linear decay segments. (d) AF decay of magnetization from a specimen from gray dolomite clast center from a different clast, showing linear, southern and steep down demagnetization trajectory. Closed symbols are projections onto the horizontal plane; open symbols, onto east-west vertical plane. Parts a and b are corrected for bedding tilt, and c and d are *in-situ* directions. Tick marks on axes each represent 1 × 10^{-4} A/m intensity.

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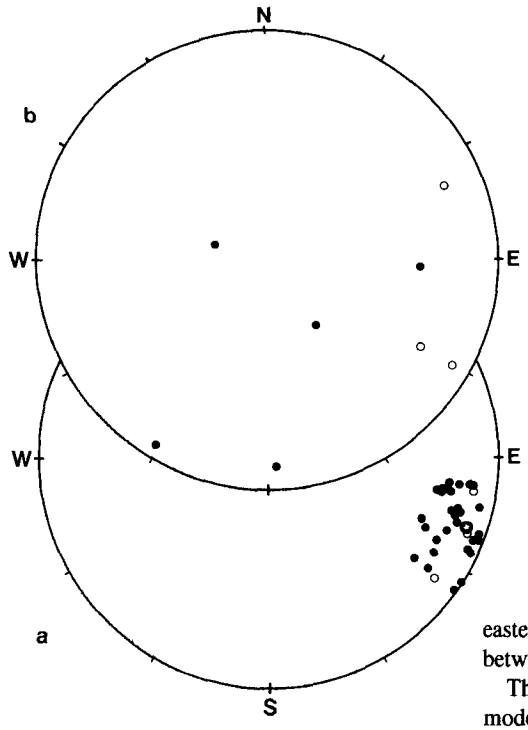


Figure 6. Stereographic projection of principal component directions. (a) Specimen directions (tilt corrected) from the gray Royer Dolomite. Mean is marked by a star. (b) Clast mean directions from gray-to-pink centers of Royer clasts in the Collings Ranch. Note wide dispersion in directions. Solid circles are projections on lower hemisphere; open circles, on upper hemisphere.

below) in evaluating the time of remanence acquisition.

Paleomagnetic Analyses

Two main components of remanent magnetization can be identified in the Royer Dolomite and correspond to the two rock types discriminated by petrographic and stable-isotope analysis. A third component is present in the karst fill and in some dolomite and dedolomite specimens.

Gray Royer Dolomite and Clast Centers. The first component is characterized by eastern and shallow magnetization directions and is found in the gray to brown Royer Dolomite that was not affected by karst processes. AF decay of the NRM was generally linear but did not always decay completely to the origin (Fig. 5a). Thermal decay of the NRM commonly removed a northern and steep component at low temperatures, interpreted to be a modern CRM residing in hematite or goethite, followed by the eastern and shallow direction at temperatures between 200 and 550 °C (Fig. 5b).

The fold test on the gray samples shows a modest increase in precision (k) parameter after correcting for bed dip (Table 1), although the grouping is only significantly different at the 63% confidence level ($k_1/k_2 = 1.2$, n = 32). The fold test, therefore, is not conclusive. The mean (tilt corrected) direction for the characteristic component from the gray Royer is Dec. = 109.2° and Inc. = 10.5° (Fig. 6a; Table 1).

Acquisition and thermal decay of IRM's indicate the presence of a low-coercivity mineral in the gray dolomites (Fig. 7a). Rapid acquisition below 0.2 T and nearly complete decay of IRM by 580 °C (Fig. 7b) suggest that magnetite is present. Because some samples continued to acquire magnetization after the initial increase below 0.2 T, high-coercivity minerals such as hematite are probably also present in these samples. Results of rock magnetic experiments and demagnetization of NRM's, however, suggest that magnetite carries the remanence.

Results of IRM acquisition and thermal demagnetization of IRM experiments on specimens from the gray to pink Royer clasts and clast centers are similar to results from the gray Royer specimens. Rapid acquisition below 0.2 T and less rapid (Fig. 7a) but significant acquisition above 0.3 T indicate the presence of both low- and high-coercivity magnetic phases. Thermal decay of partially saturated IRM's (Fig. 7b) from clast centers supports the conclusion that magnetite and some hematite are present. Figure 8 shows differences in AF decay of

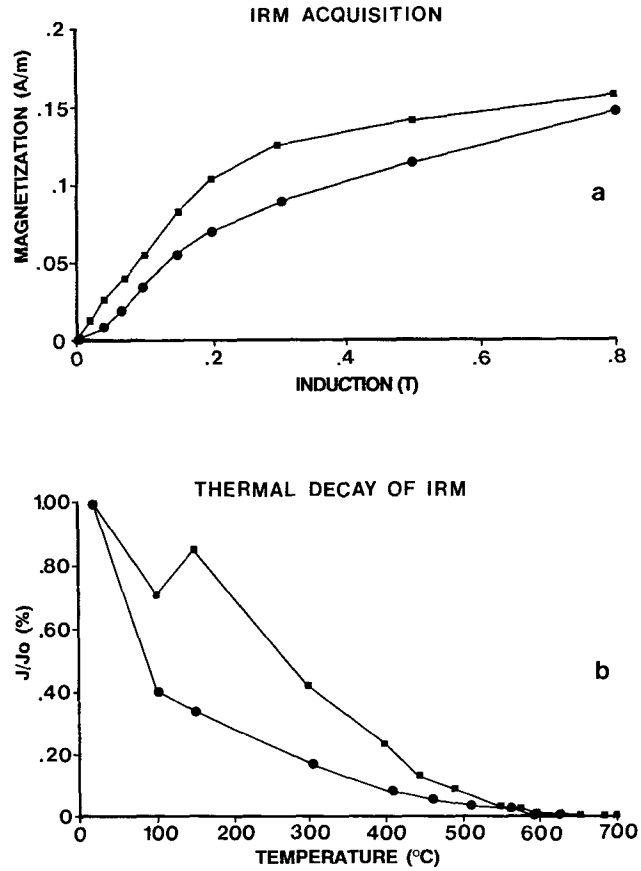


Figure 7. Rock magnetization data. (a) IRM acquisition curves for gray dolomite from Royer (squares) and centers of Royer clasts in the Collings Ranch (circles). Note rapid increase in acquired magnetization to 0.3 T and less rapid increase above 0.3 T, suggesting presence of low- and high-coercivity minerals. (b) Thermal decay of partially saturated IRM's shows 95% decay by 580 °C. A small amount of hematite may account for remaining magnetization above 580°.

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TABLE 2. SUMMARY STATISTICS OF MAGNETIZATIONS OF CLASTS OF THE ROYER DOLOMITE IN THE COLLINGS RANCH CONGLOMERATE

Clast	n/n_0	Dec.°	Inc.°	k	α_{95}°	Lat. °N	Long. °E	dp	dm
Clast means from gray clast centers									
CR 21	5/5	65.0	-16.2	351.2	4.1				
CR 30	10/12	177.7	15.2	5.5	22.8				
CR 50	3/3	202.9	5.8	22.3	26.8				
CR 60	2/2	118.8	-10.0	—	—				
CR 70	15/15	89.7	29.7	10.1	12.6				
CR 80	9/9	157.5	63.2	5.5	24.1				
CR 90	6/6	120.8	-22.8	65.8	8.3				
Mean of specimen directions from reddened rims and red clasts									
no tilt	37/50	152.5	3.7	13.6	6.6	45.5	124.0*	3.3	6.6
tilted	37/50	153.5	1.4	8.4	8.7	47.0	123.8*	4.3	8.7
Mean of clast means, red clasts and red rim directions only									
no tilt	11/14	148.5	1.4	10.6	14.7	45.5	131.0*	7.4	14.7
tilted	11/14	149.9	-3.9	9.7	15.5	47.1	130.2*	7.8	15.5

Note: see Table 1 for abbreviations.

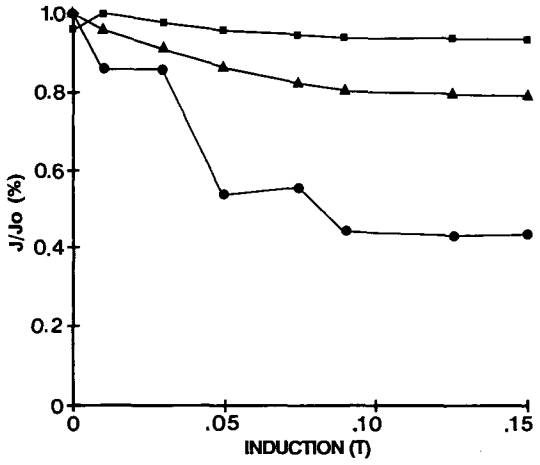


Figure 8. Response of NRM to AF treatment in three specimens from a single clast with abundant hematite. Squares mark decay curve from reddened rim and show less than 10% decrease. Circles mark decay curve from the clast center, showing decay to about 45% by 150 mT. Triangles mark decay of a specimen near the center-rim boundary and intermediate behavior.

NRM's for specimens from the red rim, red-pink boundary, and pink to gray center of a Royer Dolomite clast in the Collings Ranch. There is no decay at the rim, partial decay at the boundary, and about 50% decay of NRM in the center of the clast. The average NRM intensities from reddened clast rims are about three times the intensity of the pink clast centers (red average = 9.42×10^{-4} A/m, standard deviation = 2.57×10^{-4}, n = 35; pink average = 3.19×10^{-4} A/m, s.d. = 4.09×10^{-5}, n = 15). The response to rock magnetic experiments and AF demagnetization suggests that magnetite is an important magnetic phase in the gray to pink clast centers.

Orthogonal projections of demagnetization data from pink to gray clasts and pink to gray centers of rimmed clasts generally show linear decay to 550 °C or 150 mT (Figs. 5c and 5d). A few specimens, however, have a low-temperature (<200 °C) northern and steep down remanent magnetization (Fig. 5c). Directions within individual clasts group well, but clast mean directions from clasts have no consistent overall direction (Fig. 6b; Table 2). A randomness test (Watson, 1956) was performed on the clast means, and the results (length of resultant vector $R_0 = 2.38$, n = 8) indicate that the mean of the clasts is not significant at the 5% level. Wide dispersion of the mean directions suggests a predepositional magnetization in the pink to gray clasts and in the clast centers.

Red Royer Dolomite and Clast Rims. A second component of magnetization is found in specimens of the reddened and dedolomitized Royer Dolomite. Orthogonal projections of the demagnetization trajectories from the karst halo show linear decay to 670 °C in a southeastern and shallow direction (Fig. 9a). The mean direction from the reddened halo with no tilt correction is Dec. = 147.1°, Inc. = 3.8° (Fig. 10a; Table 1). Some specimens also contain a low-temperature component with a northern and steep direction removed by 150 °C that is interpreted to be a modern VRM or CRM.

The reddened Royer Dolomite associated with paleokarst is exposed at only one outcrop, and no fold test on this lithology was possible. The dips are shallow (10°) at this location, however, and there is little difference between tilt-corrected and *in situ* directions (Table 1).

A conglomerate test was performed on the reddened rims of Royer Dolomite clasts and a few completely red clasts from the Collings Ranch Conglomerate. Orthogonal projections of the demagnetization trajectories for specimens from clast rims show linear decay toward the origin to 670 °C (Fig. 9b). The magnetization in these specimens fails a conglomerate test because the directions from individual clasts are well grouped (Fig. 10b). Because individual clasts did not contain discernible bedding, no reorientation test comparing individual clasts was possible. A fold test on the reddened clasts was also performed. Although the test is significant only at the 63% confidence level (k_1/k_2 = 1.6, n = 37), the results suggest acquisition after folding of the conglomerate beds. The mean, *in situ* direction from the rims of the Royer clasts is Dec. = 152.5°, Inc. = 3.7° (Fig. 10b; Table 2).

IRM acquisition and decay experiments suggest that the carrier of magnetization in the specimens from the red halo around the karst and

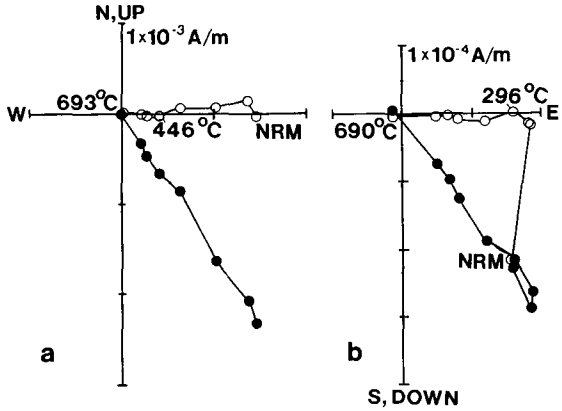


Figure 9. Orthogonal projections of specimen magnetizations during thermal demagnetization (*in situ*). (a) Reddened karst halo specimen from Royer Dolomite, showing removal of a southeastern and shallow component. (b) Reddened clast rim of Royer Dolomite in the Collings Ranch Conglomerate, showing removal of a northern and steep component followed by a southeastern and shallow component. Closed symbols are projections onto horizontal plane; open symbols, onto east-west vertical plane.

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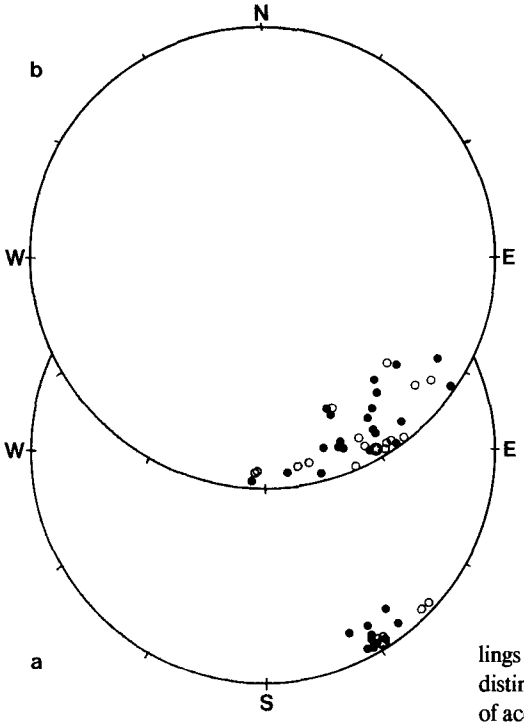


Figure 10. Stereographic projections of principal component magnetization directions from (a) the reddened karst halo and (b) red Royer clasts and reddened clast rims in the Collings Ranch Conglomerate. Mean directions are marked by stars. Solid circles are projections on lower hemisphere; open circles, on upper hemisphere.

the red clasts is a high-coercivity magnetic phase (Fig. 11a) with maximum unblocking temperatures above 580 °C (Fig. 11b). AF demagnetization results, implying a median destructive induction greater than 150 mT, also supports this interpretation (Fig. 8). These data, the high demagnetization temperatures, and the petrographic studies strongly suggest that this carrier is hematite.

The third component, a modern VRM or CRM, is found in clastic debris filling the karst cave and is also present in a few specimens of the Royer Dolomite. Thermal demagnetization removes a component with a northern and steep positive direction (Fig. 12; Table 1) at moderate unblocking temperatures (300–400 °C). Northern and steep directions were also removed at low unblocking temperatures (<200 °C) in some specimens from the Royer Dolomite.

Interpretation

The paleomagnetic results for the Royer Dolomite, supported by the results from the Collings Ranch Conglomerate, indicate at least two distinct components of magnetization. The times of acquisition are constrained by a conglomerate test and supported by the interpretation of the diagenetic history based on sedimentologic data.

The distribution of mean directions from the pink to gray clasts and the clast centers in the Collings Ranch Conglomerate indicates a predepositional magnetization. The clast centers can be correlated with the gray Royer Dolomite based on their petrographic, isotopic, and rock magnetic characteristics. Although the rock magnetic results support the use of Royer clasts in the conglomerate test, the clast centers also have some high-coercivity minerals, based on the IRM acquisition results (Fig. 7a). Directions from the clast centers may be partially contaminated by the addition of high-coercivity components. Comparison of AF and thermal demagnetization trajectories, however, supports the dominance of the low-coercivity component in the NRM (Figs. 5c and 5d). The presence of a low-coercivity carrier is further supported by the differences in response to AF demagnetization between specimens from the clast margins and centers (Fig. 8).

The eastern and shallow magnetization direction from the gray Royer Dolomite corresponds to a pole position at lat. = 12.6°N, long. = 157.3°E after tilt correction (Fig. 13; Table 1). This pole is similar to poles interpreted to be of early Paleozoic age (for example, Van der Voo and others, 1976; Watts and others, 1980; Dunn and Elmore, 1985; Jackson and Van der Voo, 1985; Loucks and Elmore, 1986) and generally supports the interpretation of the position of

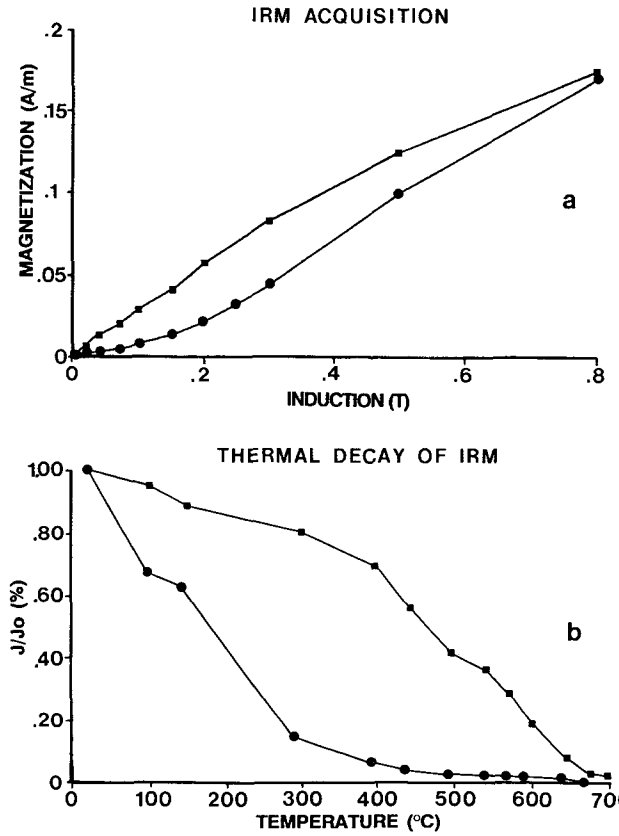


Figure 11. IRM acquisition (a) and thermal demagnetization curves (b) for samples from karst halo (squares) and reddened clast rims (circles). (a) Note gradual increase in acquired magnetization to 0.8 T, indicating the presence of high-coercivity minerals. (b) Thermal decay continues above 580 °C, suggesting that hematite is the carrier mineral.

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North America through the early Paleozoic presented by Van der Voo (1989).

The character of the magnetization of the gray Royer dolomite is quite distinct from that of reddened and dedolomitized rock. Low coercivity and maximum unblocking temperatures less than 580 °C suggest that magnetite is the carrier mineral. The origin of this magnetization in the gray dolomite cannot be positively determined. The presence of magnetite spheres supports a CRM, but a DRM cannot be conclusively ruled out.

Magnetizations from the reddened Royer Dolomite around the karst feature are southeastern and shallow and correspond to an apparent pole position at lat. = 42.3°N, long. = 130.1°E (Fig. 13; Table 1). This pole position falls on the late Paleozoic (Pennsylvanian-Permian) part of the apparent polar wander path for North America (Irving and Irving, 1982). These results are also constrained by a conglomerate test from the Collings Ranch Conglomerate, which indicates a postdepositional acquisition of remanence. The completely red clasts and the red clast rims are similar in lithology, rock magnetic properties, and remanence direction to the halo around the karst. With no tilt correction, the reddened clasts and clast rims give a pole position of lat. = 45.5°N, long. = 124.0°E (Table 2; Fig. 13).

Independent support of the age of magnetization comes from the stratigraphic and sedimentologic framework. The magnetization in the karst halos, reddened clasts, and reddened rims of clasts is interpreted to be a CRM resulting from precipitation of iron oxides during dedolomitization. Alteration was due to surficial weathering promoted by meteoric waters during late Pennsylvanian-Permian uplift.

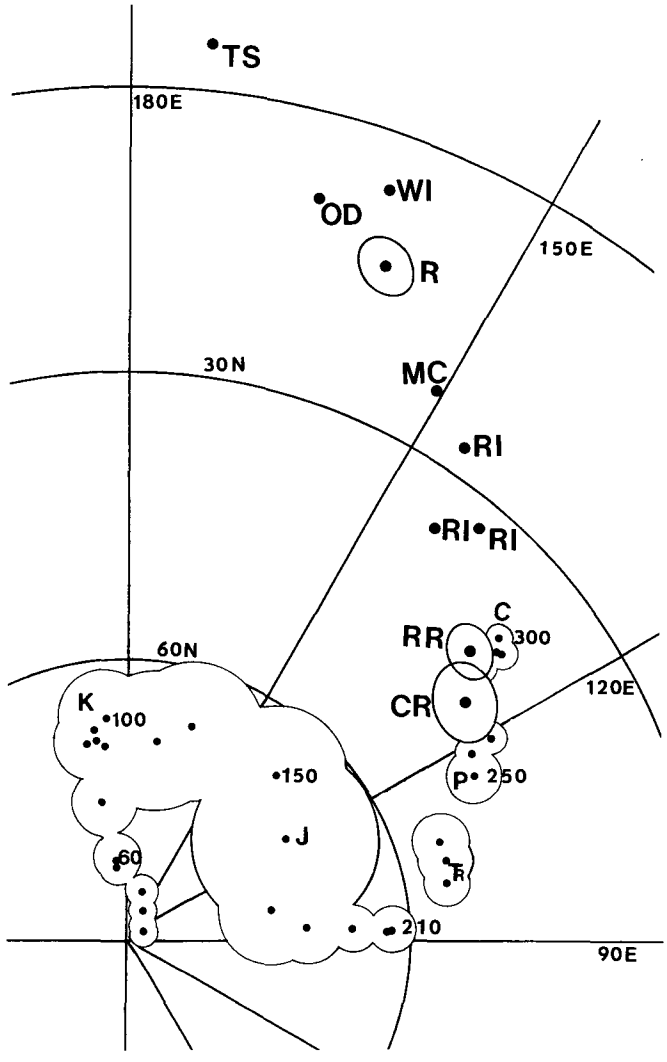


Figure 13. Apparent polar wander path for North America (after Irving and Irving, 1982) with poles and associated ovals of 95% confidence calculated from reddened karst halo (RR), red clast rims in Collings Ranch Conglomerate (CR), and the gray Royer Dolomite (R). Other poles, interpreted to be early Paleozoic in age, are shown for comparison: WI, Cambrian Wilberns Formation (Van der Voo and others, 1976; Watts and others, 1980); RI, Cambrian Riley Formation, and MC, Morgan Creek (Watts and others, 1980); TS, Cambrian Bonneterre Formation (Dunn and Elmore, 1985); OD, Ordovician Oneota Dolomite (Jackson and Van der Voo, 1985). Circles around 10 Ma poles of APWP correspond to standard errors.

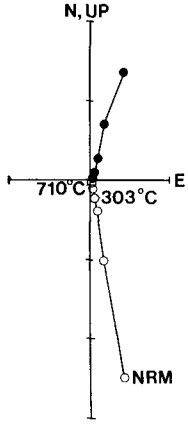


Figure 12. Projection of thermal demagnetization response of a representative specimen of karst fill material. A northern and steep down component is removed by 350 °C. Closed symbols are projections onto horizontal plane; open symbols, onto a vertical plane. Tick marks on axes each represent 1×10^{-3} A/m intensity.

DISCUSSION

The identification of high-quality reference poles for the Paleozoic has commonly been hampered by the possibility or demonstration of remagnetization. Remagnetization during the Late Pennsylvanian–Permian is common in the midcontinent region, which has led to the suggestion of regional-scale mechanisms to explain this process (for example, Oliver, 1986). As previously mentioned, several causes of remagnetization, including weathering, migration of fluids, and cooling from elevated burial temperatures, have been invoked. These mechanisms may be linked by orogenic activity and uplift. For example, the Alleghanian-Ouachita-Marathon orogeny could have resulted in uplift of deeply buried rocks, fluid migration, and enhanced weathering.

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The preservation of two magnetizations in the Royer and in clasts in the Collings Ranch Conglomerate demonstrates that weathering is an important, but nonpervasive, end-member mechanism for remagnetization. Although the effects of weathering would have been laterally widespread, they did not affect the Royer to a great depth. Older magnetizations in magnetite were apparently unaffected by any remagnetization process in the late Paleozoic.

The origin of the magnetization in the gray Royer dolomite is not obvious. Both early and late secondary magnetizations have been reported from dolomite units. For example, magnetite of biogenic origin has been identified in late Neogene–Quaternary shelf sequences, and apparently detrital remanence directions were maintained through early dolomitization (McNeil and others, 1988). Jackson and Van der Voo (1985) also reported an early magnetization in the Oneota Dolomite. Other studies have demonstrated a remagnetization apparently related to dolomitization (Addison and others, 1985; Hart and Fuller, 1988). It is not known if the magnetization in the gray Royer Dolomite was reset by dolomitization, but in any case, the remanence is an early Paleozoic DRM or CRM that has been preserved through uplift and deformation of the Arbuckle Mountains.

With respect to the origin of remagnetization, petrographic results show that hematite in the rocks is associated with dedolomitized areas and is not present in the unreddened ferroandolomite host rock. Stratigraphic, sedimentologic, and geochemical arguments all support near-surface alteration by meteoric fluids as being responsible for the remagnetization.

CONCLUSIONS

The Royer Dolomite contains a magnetization that resides in magnetite and is either a DRM or CRM. A conglomerate test on clasts of the Royer in the Collings Ranch Conglomerate indicates that the magnetization was acquired prior to the Middle to Late Pennsylvanian. The pole position for the Royer Dolomite is similar to that of other poles inferred to be early Paleozoic in age, which suggests that the magnetization was acquired in the early Paleozoic.

Reddening and precipitation of hematite occurred around and below karst features in the Royer Dolomite during dedolomitization by surface weathering in the Late Pennsylvanian to Early Permian. The early Paleozoic component residing in magnetite was replaced by a secondary remanent magnetization residing in hematite as a result of dedolomitization. The centers of Royer clasts in the Collings Ranch Conglomerate contain a predepositional magnetization that predominantly resides in magnetite and probably corresponds to the early Paleozoic component in the Royer. The dedolomitized rims of the Royer clasts contain a postdepositional component residing in hematite, which is similar to the dedolomitized Royer component.

This study demonstrates the effect of one "remagnetization" end-member mechanism, surficial weathering. A nonpervasive, Kiaman remagnetization is related to a specific diagenetic event, dedolomitization. Dolomites below the weathering zone were not remagnetized in the late Paleozoic but retain an apparent early Paleozoic magnetization.

ACKNOWLEDGMENTS

Support for this research was provided by a grant from the National Science Foundation (EAR86-17497). The authors thank Mike Engel and Rick Maynard for use of the stable-isotope laboratory and their help in running the analyses. The authors also thank M. Engel, N. Donovan, and the Geological Society of America reviewers for their helpful comments.

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